

Seven Daily Profiles, Three, or None? Weekday, Day Type and Weather in the Daily Load Shape of U.S. Balancing Authorities

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Abstract—Anomaly detection, gap repair and short-term forecasting of electricity load all rely on a notion of the typical daily profile, and practice differs on how many profiles to keep: one per weekday, or one per day type (weekday, Saturday, Sunday), each with a list of special days. This note compares the two, and several alternatives, on the public EIA-930 record of 44 U.S. balancing authorities (BAs) from 1 July 2015 to 30 June 2026 (172,741 BA-days). The daily shape is defined as the 24 hourly demand values divided by the daily mean, in local time; profiles are causal medians over a reference set of earlier days, and models differ only in how that set is chosen. Seven weekday profiles have a higher out-of-sample shape error than three day-type profiles in every BA (mean relative loss 4.3%), because mid-week profiles differ by 0.06 units of day-to-day noise while the reference set shrinks fivefold. The length of the reference window matters more than any calendar split: error rises monotonically from 3.76% at two weeks to 5.48% at sixteen. Fixed seasonal classes are 11% worse than two recent weeks, whereas adding the same three calendar weeks of the previous year to the recent two reduces error by 3.7% (confidence interval above zero in 37 BAs). The largest effect is not calendar but weather: the days on which the most BAs depart from their profiles are heating/cooling regime transitions rather than holidays, and a three-class day type from the daily mean temperature at one NOAA station per BA (heating below 59°F, cooling above 72°F) reduces error by 14.6%, with the interval above zero in every BA; the previous day’s temperature performs almost as well (12.8%), so no forecast is required. The recommended configuration, three day types crossed with the temperature regime and a reference set of the last two weeks plus the previous year’s analogs, reaches 3.22% against 3.47% for the plain three-type model. Thanksgiving, Christmas and the two eves resemble Saturdays; Memorial Day, Labor Day and Independence Day resemble Sundays; four minor federal holidays are not distinguishable from ordinary weekdays. Code and results are open.

Index Terms—load profiles, load forecasting, calendar effects, holidays, temperature, balancing authority, EIA-930, anomaly detection

I. INTRODUCTION

Most methods that operate on an hourly load series carry, explicitly or not, a model of the typical day. Short-term load forecasting uses calendar variables that distinguish weekdays and holidays [1], [2]; anomaly detectors compare each reading with what a normal day would show; gap-filling routines borrow the shape of equivalent days. Practice differs on how fine the calendar should be. One tradition keeps a profile for each of the

seven weekdays; another keeps three, for workdays, Saturdays and Sundays; both add a list of special days that follow no profile. The choice is rarely justified with data, and it has consequences: a finer calendar needs more history to estimate each profile, and more history means older data in a series whose shape drifts with the season.

Since July 2015, Form EIA-930 has made the hourly demand of every balancing authority (BA) in the contiguous United States public [3], so the question can be settled empirically and at once for the whole interconnected system. This note does that. It asks three things: whether seven profiles beat three out of sample; how long the reference history should be and whether the season should enter as a class or as an analog; and which days are actually special, both from a fixed list and from the data. The purpose is descriptive and the recommendations are conditional on the evidence presented.

II. DATA

A. Source and preparation

Hourly demand (“hourly integrated values in megawatts by hour ending time” [4]) was taken from EIA’s six-month balance files for 1 July 2015 to 30 June 2026 [3], in each BA’s local time as reported in the files. A BA-day is valid when at least 23 of its 24 hours have positive demand; on the two daylight-saving transition days per year the missing hour is interpolated and the duplicated hour averaged. Days in which any hour falls below 20% or above 300% of the daily mean are removed as corrupted (631 days). A BA enters the study with at least 600 valid days and a mean demand of at least 500 MW; 44 of 58 BAs in the files qualify, giving 172,741 BA-days. EIA’s imputed values are used as reported.

B. Daily shape

The shape of BA-day d is the vector $s_d \in \mathbb{R}^{24}$ with $s_{d,h} = D_{d,h}/\bar{D}_d$, where $D_{d,h}$ is demand in hour ending h and $\bar{D}_d$ the daily mean. Dividing by the mean separates the shape of the day from its level; the level, which depends on weather and season in ways well studied elsewhere [5], is not analysed here.

C. Calendar

Weekday and day type (workday, Saturday, Sunday) are taken from the local date. The a-priori special-day list consists of U.S. federal holidays and their observed days, the day after Thanksgiving, 24 and 31 December, and Super Bowl Sunday; it covers 4.3% of BA-days. A load-derived *heating/cooling regime* is defined from the load itself: a day is a morning-peak day if its highest hour is 12:00 or earlier, which in practice is an electric-heating morning, and an afternoon/evening-peak day otherwise.

D. Temperature

One airport station is assigned to each BA, at its main load centre or the largest city of its service area (Table IV lists them). Hourly air temperature from NOAA's Integrated Surface Database, ISD-Lite files [6], is shifted to the BA's local time and aggregated to a daily mean, minimum and maximum; days with fewer than 18 observations are dropped. A temperature regime classifies each day as heating when the daily mean is below 59°F (15°C), cooling when above 72°F (22°C), and mild otherwise. The 59°F threshold is the value that best separates morning-peak days from the rest across all BAs, and the best per-BA value lies between 55 and 63°F in every BA with a material share of morning-peak days; temperature and the morning-peak flag agree on 76% of days.

III. METHODS

A. Causal profile models

For each day d and each model, the profile $\hat{s}_d$ is the hour-by-hour median of the shapes of a reference set R_d of earlier days. Nothing after d is used, and special days are never references. The models differ only in R_d . In the *grouping* family, R_d contains the days of the same group as d in the previous W weeks, where the group is defined by one of five calendars: G1, a single group; G2, week versus weekend; G3, workday, Saturday, Sunday; G5, Monday, Tuesday–Thursday, Friday, Saturday, Sunday; G7, one group per weekday. The window W is 8 weeks for the model comparison and is varied from 2 to 16 weeks for the G1, G3 and G7 calendars. A profile is computed only when $|R_d| \geq 3$ (≥ 2 at $W = 2$).

Three further models test the season and the weekday signature. A *fixed-season* model (S4) crosses the four meteorological seasons with the three day types and uses the previous 52 weeks. A *seasonal-analog* model uses, as references, the days of the same day type within ± 21 days of the same calendar date one year earlier (A1), or one and two years earlier (A2), either alone or in union with the previous two weeks (R2+A1, R2+A2). A *hybrid* model adds to the recent G3 profile a per-weekday offset, the median over the previous 52 weeks of the residual $s_j - \hat{s}_j$ for days j of the same weekday. Finally, *regime* models split G3 by the heating/cooling regime: by the day's own load-derived regime (an oracle, since it uses the day's own peak), by the previous day's load-derived regime, by the day's own temperature regime with two classes (heating or not) or three (heating, mild, cooling), and by the previous day's temperature regime. The temperature models are causal given

a temperature forecast for the day, or given only yesterday's observation in the previous-day variants. The *recommended configuration* combines the findings: day type crossed with the three-class temperature regime, with the reference set of the last two weeks plus the previous year's analogs.

B. Error and tests

The shape error of day d under a model is the mean absolute percentage error over the 24 hours,

$$e_d = \frac{100}{24} \sum_{h=1}^{24} \frac{|\hat{s}_{d,h} - s_{d,h}|}{s_{d,h}}. \quad (1)$$

Models are compared on normal (non-special) days for which every model produces a profile. For each BA, the difference in daily error between two models is summarised by its mean, a 95% bootstrap confidence interval (2000 resamples of days) and a sign test. A BA is said to *need* the finer model when the interval excludes zero and the relative gain exceeds 5%; this criterion was fixed before the data were examined.

C. Weekday distinctness

For each BA, long-run weekday profiles p_w ($w = 1, \dots, 7$) are the medians of all normal days of that weekday. The distance between two weekdays is the MAPE between their profiles, and the day-to-day noise is the median over all normal days of the MAPE between a day's shape and its own weekday profile. The ratio of the two is a dimensionless distinctness index: a value below 1 means that the two weekdays differ by less than two days of the same weekday do. Average-linkage hierarchical clustering of the index matrix, cut at 1, gives the number of weekday groups that the data distinguish.

D. Special days

For each special day and BA-year, the day's shape is compared with three causal profiles from the same 8-week window: that of its own weekday, of Saturday and of Sunday; for special days falling on a weekend only the last two apply. The closest profile is recorded. For Super Bowl Sunday, the residual against the median of the non-special Sundays within five weeks is reported by local hour and by time zone. Data-driven special days are the dates with the highest median shape error under G3 across BAs, restricted to dates on which at least 80% of BAs report.

IV. RESULTS

A. Seven profiles against three

Fig. 1 shows the mean shape error of the five grouping calendars at $W = 8$, one point per BA, and Table I the means over BAs. G7 has a higher error than G3 in every one of the 44 BAs, with the bootstrap interval entirely below zero in 44 of them; the mean relative loss is 4.3%. G5 also loses to G3 (1.8%). The distinctness index explains why (Fig. 2): the median index between Tuesday and Wednesday is 0.06, between Monday and Thursday 0.19, between Thursday and Friday 0.19, whereas between Wednesday and Sunday it is 0.62. Cutting the clustering at an index of 1 yields a single weekday

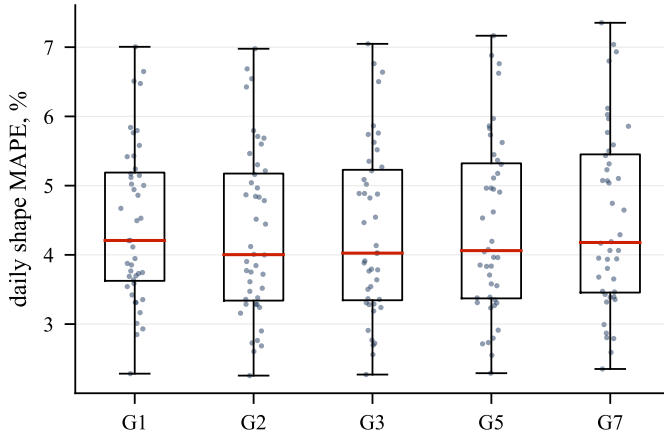


Fig. 1. Mean out-of-sample daily shape MAPE of the five grouping calendars with an 8-week reference window; one point per BA, boxes over the 44 BAs. Source: EIA-930 [3], July 2015 to June 2026.

TABLE I
MEAN DAILY SHAPE MAPE OVER BAS, NORMAL DAYS, JULY 2015 TO JUNE 2026.

Model	Reference set (day type $\times$ history)	MAPE
G1	one group, 8 weeks	4.45%
G2	week / weekend, 8 weeks	4.31%
G3	weekday / Sat. / Sun., 8 weeks	4.34%
G5	Mon. / Tue.–Thu. / Fri. / Sat. / Sun., 8 weeks	4.40%
G7	one per weekday, 8 weeks	4.51%

group in 37 BAs and two groups (workdays; Saturday and Sunday) in 7: CISO, LDWP, NYIS, PGE, PSEI, SCL, TPWR. The hybrid model confirms that the weekday signature is real but immaterial: it improves Mondays by 0.05 and Fridays by 0.08 percentage points, costs the same on the other days, and changes the overall error by -0.3% relative.

B. Reference window and season

Fig. 3 shows the error of G1, G3 and G7 against the window length. Error rises monotonically with W for every calendar: G3 goes from 3.76% at two weeks to 5.48% at sixteen, and a single profile from the last three weeks (3.86%) beats three profiles from the last six (4.07%). The best model in the grid is G3 at $W = 2$ weeks (3.76%).

The season enters usefully as an analog rather than as a class (Table II). The fixed-season model S4 is 11% worse than two recent weeks, with the interval below zero in 39 BAs. The analogs of the previous year alone are as accurate as the last two weeks (3.81% against 3.79% on the same days); their union with the recent two weeks is better than either (3.63%), a relative gain of 3.7% with the interval above zero in 37 BAs and above 5% in 25; two years of analogs give 3.9%. The gain is present in every month and largest from March to June (0.19 against 0.11 percentage points in the other months). The exceptions are WALC, CISO and WACM, where the analogs increase error because the daily shape of those systems is changing from one year to the next.

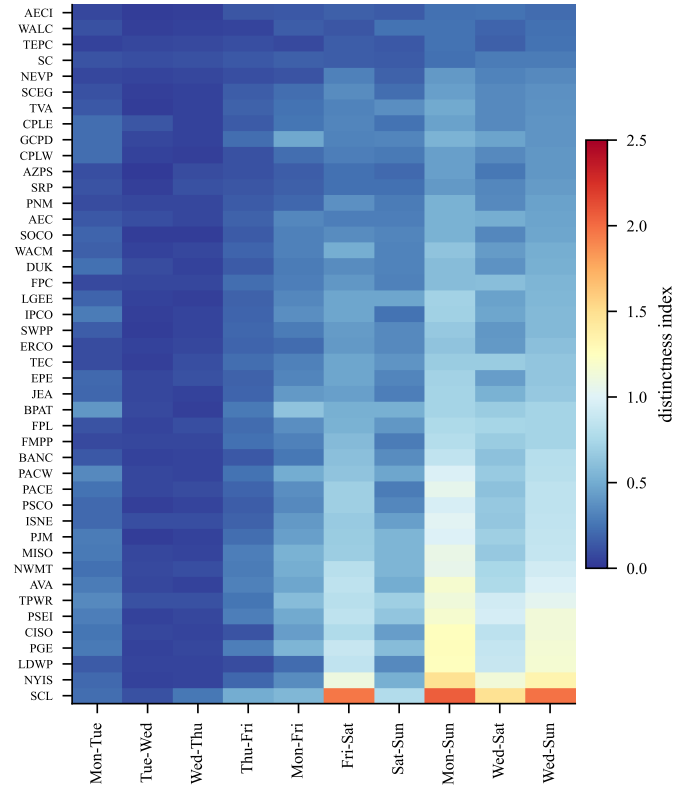


Fig. 2. Distinctness index between pairs of weekdays, one row per BA, sorted by the Wednesday–Sunday index. An index of 1 means that the two weekday profiles differ as much as two days of the same weekday do.

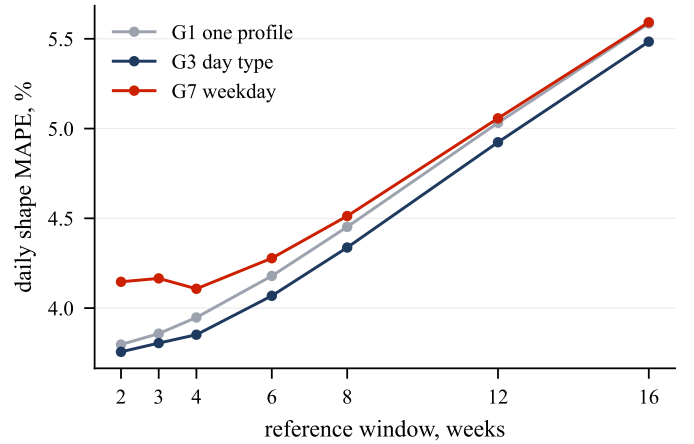


Fig. 3. Mean daily shape MAPE over BAs against the length of the reference window, for the G1, G3 and G7 calendars.

C. Heating/cooling regime

Fig. 5 shows, on the left, the monthly share of morning-peak days for five BAs, and on the right the gain of the load-derived regime split over G3 by BA. In the Southeast and the Northwest more than 40% of December and January days peak in the morning; in California, New York and New England almost none do. Splitting by the day's own load-derived regime reduces the shape error by 10.0% on average and by up to 19% in

TABLE II

SEASON AS A CLASS VERSUS AS AN ANALOG. MEAN DAILY SHAPE MAPE OVER BAS ON THE DAYS WHERE ALL FIVE MODELS EXIST; RELATIVE GAIN OVER THE TWO-WEEK G3 MODEL WITH THE NUMBER OF BAS WHOSE 95% INTERVAL LIES ABOVE (OR BELOW) ZERO.

Model	Reference set	MAPE	Gain (BAs +/-)
R2	weekday/Sat./Sun., prev. 2 wk	3.79%	—
S4	season $\times$ day type, prev. 52 wk	4.19%	−11.1% (5/39)
A1	analogs ± 21 d, prev. year	3.81%	—
R2+A1	$R2 \cup A1$	3.63%	+3.7% (37/3)
R2+A2	$R2 \cup$ two years of analogs	3.62%	+3.9%

TABLE III

REGIME MODELS: MEAN DAILY SHAPE MAPE OVER BAS ON THE DAYS WHERE ALL MODELS EXIST (8-WEEK WINDOW), RELATIVE GAIN OVER G3, AND NUMBER OF BAS WITH THE 95% INTERVAL ABOVE ZERO.

Model	Regime	MAPE	Gain	BAs
G3	none	3.97%	—	—
G3R	peak hour, same day	3.53%	10.0%	42
G3Rprev	peak hour, previous day	3.80%	4.4%	40
G3T	temp., 2 classes, same day	3.64%	8.6%	44
G3T3	temp., 3 classes, same day	3.37%	14.6%	44
G3Tprev	temp., 2 classes, prev. day	3.63%	8.7%	44
G3T3prev	temp., 3 classes, prev. day	3.44%	12.8%	44

AEC; the previous-day proxy gives 4.4%. The dates on which the most BAs depart from their profiles at once (Table V) are regime transitions, the first cold morning of autumn and the first hot afternoon of spring, rather than holidays; the highest-ranked holiday, Thanksgiving, appears at rank 13.

D. Temperature regime

Fig. 4 shows the share of morning-peak days against daily mean temperature (left) and the gains of the temperature-based splits by BA (right); Table III summarises the regime models. The share of morning-peak days falls from above 60% below 40°F to zero above 68°F in every BA with electric heating. The two-class temperature split reduces the shape error by 8.6%, with the interval above zero in all 44 BAs; the three-class split reduces it by 14.6%, above 5% in 41 BAs, and outperforms the load-derived oracle. The previous day’s temperature is nearly as informative as the day’s own: 8.7% with two classes and 12.8% with three. The recommended configuration reaches 3.22% on the days where all models exist, against 3.47% for the two-week three-type model, a reduction of 10.3% with the interval above zero in 42 BAs; with the previous day’s temperature it reaches 3.28%.

E. Special days

Table VI gives, for each special day, the median error against the three candidate profiles and the profile most often closest. Three groups appear. Thanksgiving, Christmas Day, Christmas Eve and New Year’s Eve are closest to Saturday in a majority of BA-years; Memorial Day, Labor Day and Independence Day are closest to Sunday; MLK Day, Presidents’ Day, Columbus Day and Veterans Day are closest to their own weekday, i.e. they are not special days for load. Juneteenth, observed federally only since 2021, is split. Super Bowl Sunday shows a residual

TABLE IV
REFERENCE STATIONS (NOAA ISD).

BA	Station	BA	Station	BA	Station
AEC	KMGH	GCPD	KMWH	PSCO	KDEN
AECI	KSGF	IPCO	KBOI	PSEI	KSEA
AVA	KGEG	ISNE	KBOS	SC	KCHS
AZPS	KPHX	JEA	KJAX	SCEG	KCAE
BANC	KSMF	LDWP	KLAX	SCL	KSEA
BPAT	KPDX	LGEE	KSDF	SOCO	KATL
CISO	KLAX	MISO	KIND	SRP	KPHX
CPL	KRDU	NEVP	KLAS	SWPP	KOKC
CPLW	KAVL	NWMT	KBIL	TEC	KTPA
DUK	KCLT	NYIS	KJFK	TEPC	KTUS
EPE	KELP	PACE	KSLC	TPWR	KSEA
ERCO	KDFW	PACW	KPDX	TVA	KBNA
FMPP	KMCO	PGE	KPDX	WACM	KDEN
FPC	KMCO	PJM	KPHL	WALC	KPHX
FPL	KMIA	PNM	KABQ		

TABLE V

DATES WITH THE LARGEST MEDIAN SHAPE ERROR ACROSS BAS (G3, 8 WEEKS), AMONG DATES ON WHICH AT LEAST 80% OF BAS REPORT.

Date	Weekday	Median MAPE	Special day
01 Nov 2019	Fri	8.6%	—
01 Nov 2023	Wed	8.2%	—
19 Nov 2022	Sat	7.9%	—
14 May 2018	Mon	7.9%	—
13 Jun 2021	Sun	7.8%	—
07 May 2018	Mon	7.8%	—
06 Jun 2016	Mon	7.7%	—
19 Oct 2024	Sat	7.5%	—
13 May 2023	Sat	7.5%	—
03 Jun 2020	Wed	7.4%	—
05 Jun 2016	Sun	7.3%	—
02 Nov 2019	Sat	7.3%	—

of −3 to −4% of the daily mean during the game, shifted by time zone (Fig. 6), and is closer to a Saturday than to a Sunday in 61% of BA-years.

V. DISCUSSION

A. Interpretation

The results separate three effects that are usually bundled under “calendar”. The weekend effect is real in every BA and dominant in the urban and West-Coast systems. The weekday effect within the working week exists, is stable, and is smaller than the day-to-day noise; estimating it costs more history than it returns, whether by grouping or by a long-run offset. The seasonal effect is the largest of the three and is best handled by recency, with the previous year’s analog as a complement that anticipates where the shape is going, and worst handled by fixed classes. Above all three sits the weather regime: on the days that matter most for operations, when the peak moves from the evening to the morning, no calendar model is informed, and a temperature-conditioned day type recovers more error than every calendar distinction combined. That the previous day’s temperature performs almost as well as the day’s own means the regime can be used without a forecast, because the regime persists from one day to the next far more often than it switches. That the largest coordinated departures from the typical shape are regime transitions rather than holidays is, for anomaly detection, the central finding: a detector that compares

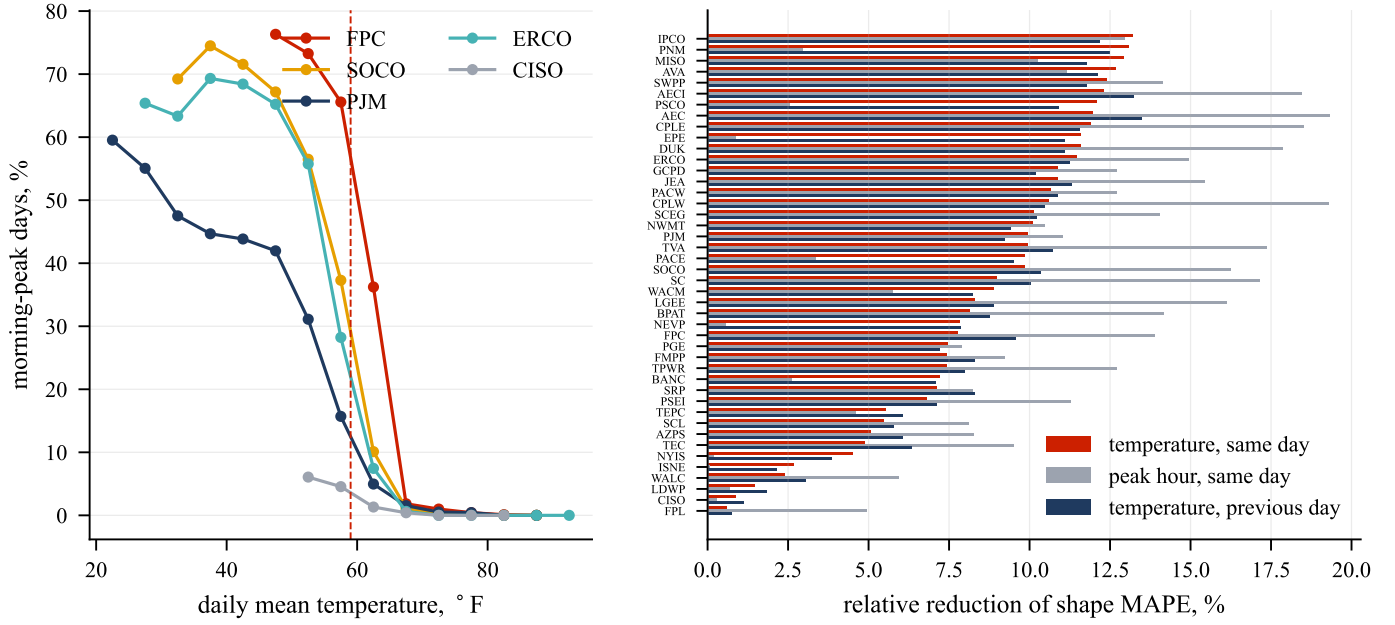


Fig. 4. Left: share of morning-peak days against daily mean temperature at the reference station, for five BAs; the dashed line is the 59°F threshold. Right: relative reduction of shape MAPE over G3 from the two-class temperature split (same day and previous day) and from the load-derived same-day regime, by BA.

TABLE VI

SPECIAL DAYS: MEDIAN SHAPE MAPE AGAINST THE PROFILE OF THE DAY'S OWN WEEKDAY, OF SATURDAY AND OF SUNDAY (8-WEEK WINDOW), AND THE PROFILE MOST OFTEN CLOSEST, WITH ITS SHARE OF BA-YEARS.

Day	<i>n</i>	Own	Sat.	Sun.	Closest
Thanksgiving	476	6.5%	5.5%	6.9%	Sat. (66%)
Christmas Day	560	4.5%	3.2%	3.7%	Sat. (68%)
Christmas Eve	474	4.0%	3.1%	4.1%	Sat. (72%)
New Year's Eve	417	3.3%	2.9%	3.6%	Sat. (55%)
Day after Thanksgiving	475	4.4%	3.7%	4.3%	Sat. (47%)
New Year's Day	604	4.3%	3.5%	3.4%	Sun. (45%)
Memorial Day	474	5.9%	5.3%	4.3%	Sun. (64%)
Labor Day	472	3.7%	3.4%	2.9%	Sun. (50%)
Independence Day	511	4.6%	3.7%	3.6%	Sun. (46%)
Juneteenth	343	4.6%	4.8%	4.8%	Sun. (40%)
Super Bowl Sunday	474	—	3.4%	3.6%	Sat. (61%)
MLK Day	473	3.2%	3.4%	3.9%	own (56%)
Presidents' Day	476	3.4%	3.9%	4.0%	own (53%)
Columbus Day	474	3.8%	4.8%	5.4%	own (66%)
Veterans Day	600	4.0%	4.5%	5.5%	own (47%)

readings with a calendar profile will raise its loudest alarms on the first cold morning of the year.

B. Recommendations

For profile-based methods on U.S. BA load the evidence supports: (i) three day types rather than seven; (ii) a reference set of the last two weeks plus the same three calendar weeks of the previous year, with the analog dropped where the shape is known to be changing year over year; (iii) a special-day list restricted to New Year's Day, Memorial Day, Independence Day, Labor Day, Thanksgiving and the day after, Christmas Eve and Day, New Year's Eve and Super Bowl Sunday, each mapped to the Saturday or Sunday profile as in Table VI; and (iv) a three-class temperature regime (heating below 59°F,

cooling above 72°F of daily mean temperature) crossed with the day type, from a forecast when available and from the previous day's observation otherwise. Combined, these reduce the shape error by 10.3% relative to the plain three-type model.

C. Limitations

Only the shape is analysed; a day whose level changes without a change of shape is invisible here. One station represents each BA; for systems spanning several climates (MISO, SWPP, PJM) it is a proxy for the load centre and the temperature gains are, if anything, understated. Thresholds are stated in Fahrenheit, the unit of U.S. operational practice; the temperature thresholds are national; per-BA thresholds would fit better and generalise worse, and the sensitivity is small because the best threshold is nearly the same everywhere. The 8-week window of the model comparison is not the best window, but the ranking of calendars does not change with the window. BAs spanning several time zones report in one and their profiles are blends. The special-day list is national; state holidays and local events are not represented. Results are for aggregate BA load and need not hold at the feeder or substation level, where the weekday signature can be larger.

VI. CONCLUSION

On the public record of 44 U.S. balancing authorities, three day-type profiles estimated from recent days outperform seven weekday profiles everywhere, the length and recency of the reference history matter more than the calendar, the previous year's analog days add to recent days, and the heating/cooling regime moves the daily shape more than any holiday. A short list of special days, mapped to Saturday or Sunday profiles, covers the calendar exceptions that the data support. These

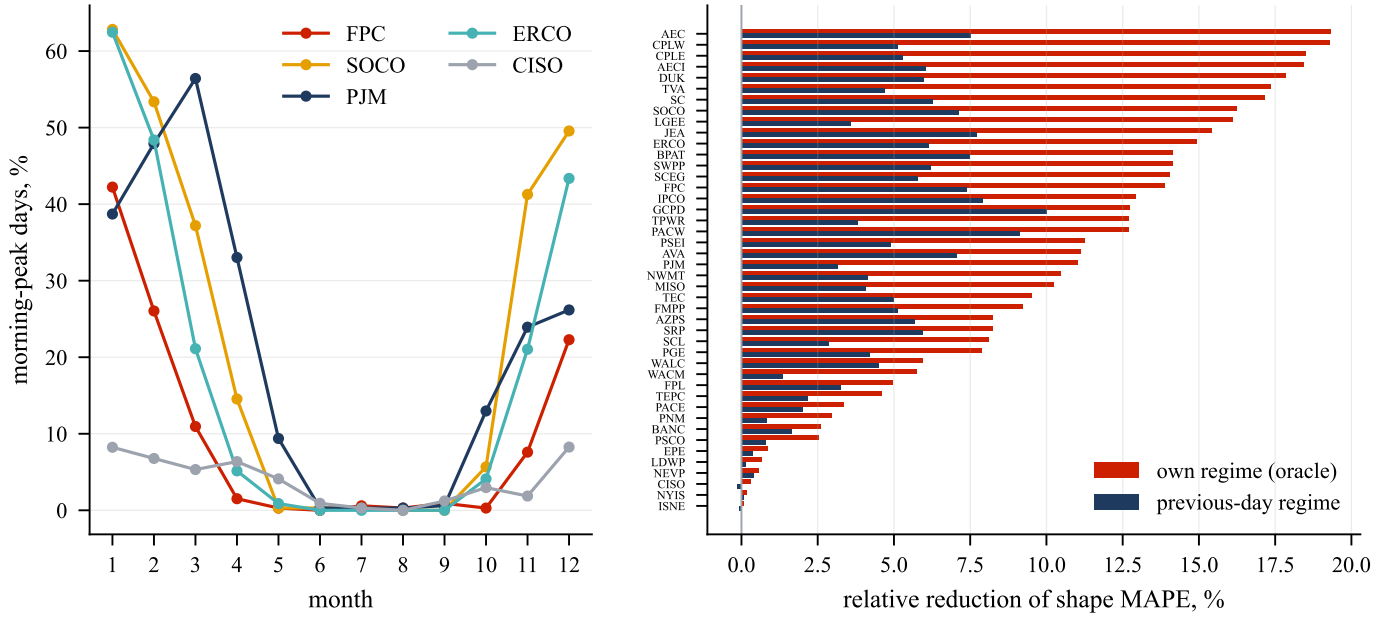


Fig. 5. Left: share of days whose peak falls at hour ending 12 or earlier, by month, for five BAs. Right: relative reduction of shape MAPE from splitting the three day types by the heating/cooling regime, using the day's own regime (oracle) and the previous day's regime (causal), by BA.

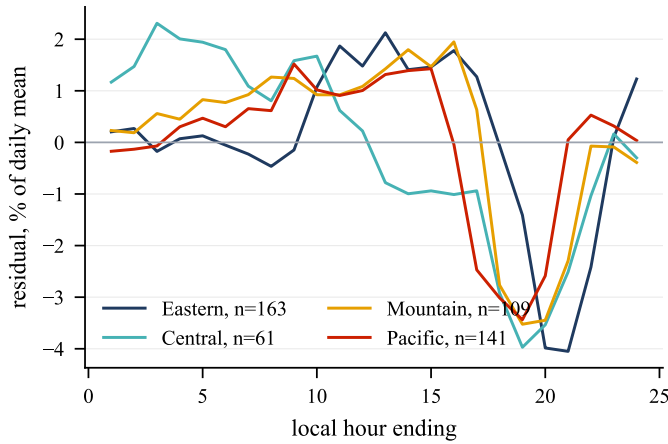


Fig. 6. Super Bowl Sunday: median shape residual against neighbouring Sundays, by local hour and time zone; n is the number of BA-years.

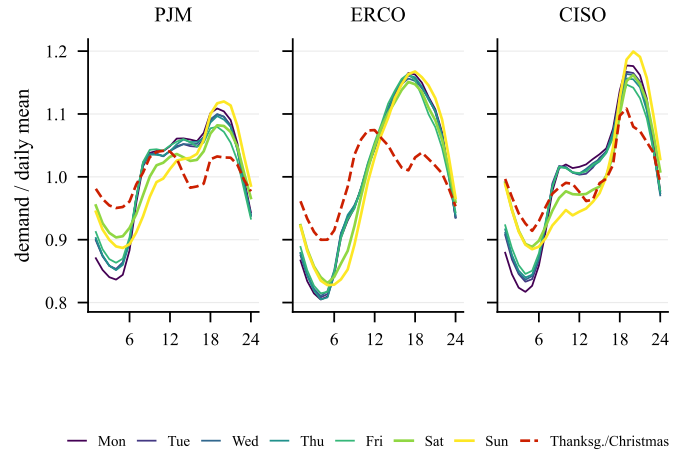


Fig. 7. Long-run weekday profiles of three BAs (medians over all normal days) with the median Thanksgiving and Christmas Day shape overlaid.

choices are being adopted in the open anomaly-detection tool of the same initiative.

DATA AND CODE AVAILABILITY

Data: EIA-930 six-month balance files [3], prepared with [7]; NOAA ISD-Lite hourly observations [6]. This note, its figures and tables are reproduced by `analysis.py` and `paper.py` in <https://github.com/cardinalgrid/notes> (code Apache-2.0, text and figures CC BY 4.0).

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//github.com/cardinalgrid/ba-forecast-scorecard.